Message Drives Image: A Coverless Image Steganography Framework Using Multi-Domain Image Translation

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2021 International Joint Conference on Neural Networks (IJCNN) | 978-1-6654-3900-8/21/$31.00 ©2021 IEEE | DOI: 10.1109/IJCNN52387.2021.9534043

***Abstract*—The coverless steganography methods based on the GAN satisfy the need of large image set which is a shortcoming for traditional coverless methods. But most of GAN based methods have either low hiding capacity or weak robustness. In this paper, to solve these problems we design a coverless image steganography framework based on multi-domain image translation called MDI. The target label used to translation is built a relationship with secret message. This framework includes a generator and a multi-label classifier. Generator is used to translate cover image into stego image according to target label. Classifier is adopted to extract label in the image and obtain secret message. Moreover, common types of image attacks are considered in advance to improve the robustness of classifier. Finally, our experiments and discussion show our method has more robustness and higher hiding capacity than state of art coverless image steganography methods.**

I. Introduction

Steganography, which embeds a secret message into a cover object with a hidden manner [1], is applied in the public channel to ensure that the existence of communication process is not detected. The schema of steganography model is shown in Fig. 1. We use Alice and Bob to represent sender and receiver respectively. Firstly Alice embeds the secret message into cover object and obtains stego object which represents cover with secret message. Then Alice transmits stego or cover object in the public channel. There is a steganalyzer in the public channel aiming to distinguish if there is a secret message in the cover. What's more, it's not necessary for steganalyzer to reveal the contents of secret message. Finally, Bob receives stego and recovers secret message. Specificly, image steganography means that Alice and Bob can hide secret message in cover image and transmit stego image in the public channel without causing the suspicion of steganalyzer.

In image steganography, most traditional embedded methods embed secret message into cover image as little distortion as possible to minimize the detectability [2]. But at the same time, more and more advanced steganalysis algorithms are being proposed constantly. In order to reduce the risk of being detected by steganalyzer, a novel concept called coverless image steganography is proposed.

*Rui XUE is the corresponding author of this paper. This research was supported by the National Key R&D Program of China under Grant No. 2019QY(Y)0602, National Nature Science Foundation of China under Grant No. U1936119 and National Nature Science Foundation of China under Grant No. 61941116.

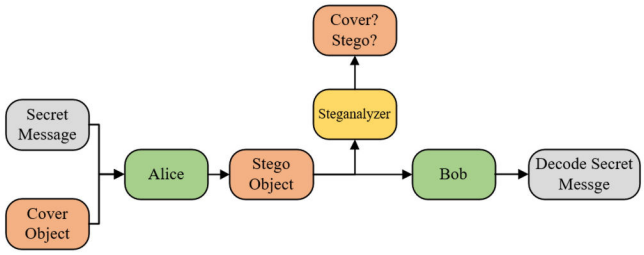

Fig. 1. The schema of steganography.

Coverless does not mean there is no cover image during image steganography. Actually, it means using cover image without making any modification. In other words, coverless methods usually use natural images own rich information [3] [4] [5] or use generative adversarial network (GAN) [6] to generate image driven by secret message [7] [8] and realize image steganography.

Since no modification traces will be left, coverless image steganography can resist existing steganalysis tools. In Zhou et al. [3], a hashing algorithm is used to index the images collected by themselves. The message is composed of the index. Therefore, there is a relationship between image and secret message. Zheng et al. [4] use the orientation information of the scale-invariant feature transform (SIFT) feature points to resist common image attacks, such as rotation with a certain angle and gaussian noise. Yuan et al. [9] use SIFT, bag of features (BOF) and hash function to represent secret message. It has a good robustness to resist common image attacks but the hiding capacity is low. Most of above traditional coverless methods are not practical because of the large number of image dataset that they need to represent all the secret message, which can be evaluated by coverage rate [5]. In contrast, we want to propose a coverless framework that only needs small-scale number of cover images but can reach 100% coverage rate.

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GAN can generate an image from initial random noise, hence it becomes a popular strategy for coverless image steganography methods to solve above problem. Hu et al. [7] utilize DCGAN [10] to generate stego image, and predefine the relationship between noise vector and secret message. Cao et al. [8] take advantage of DRAGAN [11] and Illustration2vec [12] to generate anime characters and reveal labels. A concurrent work proposed by Bi et al. [13] uses the generator and classifier of StarGAN [14] to produce stego image and classify stego image. The label in the image is secret message. It has a fine accuracy result in the condition of a low hiding capacity. Unfortunately almost all the existing GAN based method has the following disadvantages: the visual effect is not stable, the model is not robust under image attacks in the public channel, such as gaussian noise and median filter, and some of them have low hiding capacity. Thus, we propose a robust, good visual effect and high hiding capacity coverless framework here.

In this paper, a coverless steganography framework based on multi-domain image translation, called MDI (the abbreviation of **m**essage **d**rives **i**mage), is designed by us. MDI can establish a specific relationship between secret message and high-level semantics, i.e. attributes label, of image to realize coverless image steganography. It takes advantage of GAN to address the shortage of cover image. MDI adopts multi-domain image translation to control attributes as much as possible, which is beneficial to increase hiding capacity and raise efficiency. In order to extract secret message correctly from attacked image to the most extent and keep framework robust, MDI utilizes an extra multi-label classification network with data augmentation to extract secret message. The whole framework consists of a generator trained in GAN and a multi-label classifier based on residual network. The secret message is divided into segments and mapped into target labels. Then the target label and cover image are used as the input of generator to obtain stego image. The label of stego image is extracted by classifier and inverse map into secret message.

Our framework can realize 100% coverage rate [5] and have good robustness. Thanks to multi-domain image translation, MDI can only use a single network to translate image according to multiple attributes accurately. And once secret message is input into the framework, MDI can automatically generate image having corresponding attributes. So it is unnecessary for Alice to care about which image to be sent. In summary, the main contributions of our paper are as follows:

- A novel framework using multi-domain image translation for coverless steganography is proposed. A generator is applied in our framework to hide secret message and generate stego images. A multi-label classifier is used to extract secret message from the stego images.
- Multi-domain image translation based on GAN is adopted. Through this way, we realize MDI that reaches 100% coverage rate, overcomes the problem that large-scale cover image set is needed in the traditional coverless steganography method and has a higher hiding capacity.
- An extra multi-label classification network robust against arbitrary types of image attack is trained to extract secret message. Based on the idea of data enhancement, generated images with and without attack are used as training set in order to enhance the robustness of classification network.

The remainder of this paper is organized as follows: Section II discusses related work about coverless image steganography and image translation based on GAN. Section III introduces our proposed framework. The experiments and discussion are shown in Section IV. Section V is used to conclude this paper.

II. Related Work

A. Traditional Embedded Image Steganography Methods

In steganography field, traditional embedded methods realizes steganography based on spatial domain [15] [16] and transform domain [17] [18]. In order to avoid detectable, most algorithms embed secret information into cover image with minimizing the distortion of the image [19].

However, modification during embedding secret message may be perceived in the cover image, which makes it possible for steganalyzer to detect. The steganalyzer can always use statistics or deep learning methods to distinguish stego image from cover image.

B. Traditional Coverless Image Steganography Methods

The emergence of coverless image steganography overcomes above shortcomings of image steganography. Instead of embedding secret message into spatial domain or transform domain, coverless image steganography methods use feature information, such as pixel value, color, texture, edge and high-level semantics to establish a relationship between image and secret message [20]. Owing to no modification on cover image, this method can resist the detection of steganalyzer and improve the security of image steganography.

In Zhou et al. [3], a hashing algorithm is used to index the images while the secret message is also composed of these indexes. Therefore, there is a relationship between image and secret message. When sending secret message, the sender can send the corresponding image according to the hash value that equals to the message. Yuan et al. [9] use SIFT extraction algorithm, bag of features (BOF) and hash function to represent secret message. It has good robustness to resist common image attacks, but the hiding capacity is low. Based on [3], Zheng et al. [4] designed an image hash algorithm using the orientation information of the SIFT feature points and each secret message corresponds to a certain image through hash value. But it is not practical because of the need of large number of image dataset. Wang and Wu [5] proposed a deep learning coverless steganography framework which searches for map rules between images and secret message thanks to overfitting. But as they said, the training time increases exponentially with each additional bit of secret message.

In short, most of coverless image steganography methods have a drawback that if there are n bit secret message, about 2^n natural stego images are required to represent all the secret message. It is an exponential increase for the demand of stego

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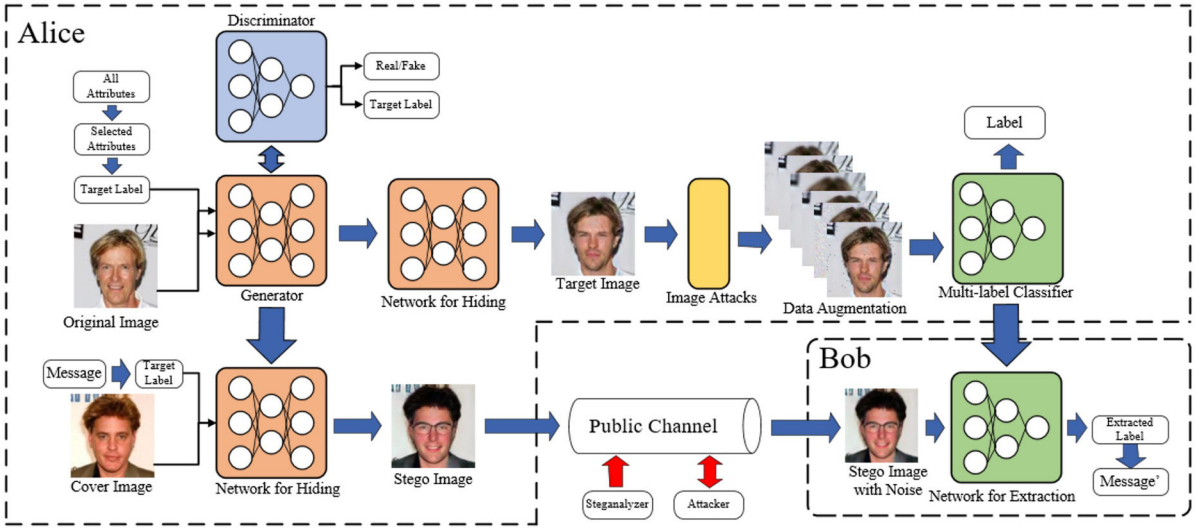

Fig. 2. The framework of MDI.

images. Hence, it is difficult for a small image dataset to present the whole secret message using traditional coverless image steganography method.

C. Coverless Image Steganography Method Based on GAN

The rise and development of GAN [6] help researchers come up with more ideas about coverless image steganography. The image generated by GAN is used as stego image, which solves the quantity shortage of cover images in traditional coverless method. Hu et al. [7] map secret message into noise vector. The stego image are generated by DCGAN [10] with specific noise vector. The receiver uses an extra extractor to recover noise from stego images and maps noise inversely into secret message. It has a higher hiding capacity but does not consider robustness. Cao et al. [8] map the attributes of anime characters images generated by DRAGAN [11] into the secret message. The receiver can use Illustration2vec [12] to recover secret message. But it's not robustness in the face of noise attack and filter attack. A concurrent work proposed by Bi et al. [13] uses the generator of StarGAN [14] to produce stego image, uses the classifier in the discriminator to classify stego image. The label in the image is mapped into secret message. It has a fine result in the condition of low hiding capacity but is ineffective facing with more attributes.

In a word, though the coverless method based on GAN can overcome the shortcoming that requires large number cover image in some extend, visual effect, the robustness and hiding capacity are still the problems to be solved.

D. Image Translation Based on GAN

The task of image-to-image translation is to change an attribute in a certain image from one to another e.g., changing hair color of a person from black to golden. Pix2pix [21] adopts L1 loss with adversarial loss and realizes image-to-image translation using paired data samples. CycleGAN [22] uses a cycle consistency loss to maintain key attributes between input and output image. In StarGAN [14], it only uses a single generator to perform multi-domains image translation. E.g. it can only use a single generator to transform a **young woman** with **black** hair (e.g. original label [1, 1, 0]) an **old man** with **gray** hair (e.g. target label [0, 0, 1]). But this process will be realized by other GANs with several subprocesses, e.g. first transform young into old, then woman into man, at last black hair into gray hair. On the other hand, StarGAN can utilize the global feature and precisely control the attribute of the image to maximize image quality.

III. THE PROPOSED COVERLESS STEGANOGRAPHY METHOD

In this paper, we propose a steganography framework based on multi-domain image translation, called MDI (the abbreviation of **m**essage **d**rives **i**mage), to realize the task of coverless image steganography, achieving the state-of-art performance.

In deed we use the definitions mentioned in [14] as follows:

- *Attribute*: a feature inherent in an image, such as gender or age.
- *Label*: a specific value of an attribute, such as male/female for gender or old/young for age.
- *Domain*: a series of images sharing the same attribute value, such as images of the aged or images of the men.

So multi-domain image translation means translating an original image into a target image among multiple domains according to a specific target label. In the whole process, there are two or more attributes in the image can be changed at the same time. And we use original image, target image and target label as cover image, stego image and the mapping of secret message separately to realize coverless image steganography in MDI.

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A. Overview

As shown in Fig. 2, there are two stages i.e. stego image generation and secret message extraction in MDI.

In the first stage, Alice selects appropriate attributes for image translation and trains the Network for Hiding (N_H). It is a generator G in GAN used to realize multi-domain image translation. After acquiring G, Alice divides secret message into fragments with a fixed length of the corresponding label. Next Alice inputs an image with its label in the dataset into N_H to get a stego image. At last Alice transmits these stego images through public channel where there are image attackers and steganalyzers.

In the second stage, Alice trains the Network for Extraction (N_E) with data augmentation and shares it with Bob through secure channel in advance. N_E is a multi-label classifier for extracting secret message from stego image. After receiving the stego images under attack, Bob uses N_E to extract label and restores secret message according to the mapping rules.

The main symbols used in the article is illustrated in Table I. Next, we will introduce our framework in detail through above two stages.

TABLE I
SYMBOLS USED IN THIS PAPER.

Symbol	Description
c	Ciphertext from secret message M
D	Discriminator network in GAN
G	Generator network in GAN
I_{cover}	The set of cover images
I_{stego}	The set of stego images
I'_{stego}	The set of stego images under attack
l	Label vectors
l'	Recovered label vectors
$Length(x)$	The function of getting the length of x
L_{attr}	The fixed length of the network input for selected attributes
M	Secret message
M'	Recovered secret message
N_E	Network for Extraction
N_H	Network for Hiding

B. Stego Image Generation

The stage of stego image generation includes four steps. In the first step, Alice evaluates if a certain attribute is suitable for translation. In the second step, Alice trains a GAN, which is used for multi-domain image translation with selected attributes, and preserves generator G as N_H. In the third step, Alice divides secret message M into M_i. M_i is encrypted with stream cipher to get ciphertext c_i, then c_i is mapped into label l_i. In the fourth step, Alice chooses a cover image I^i_{cover} and takes advantage of N_H with I^i_{cover} and l_i to obtain a stego image I^i_{stego}.

The algorithm of stego image generation is shown in Algorithm 1. Firstly, Alice evaluates which attributes are used as the candidate translation attributes. The details will be introduced later in this subsection. Then Alice trains a GAN with selected attributes and image dataset. The generator G is used as N_H. Next, according to the number of selected attributes L_{attr}, Alice gets the length of each message segment M_i. She divides secret message M into a set of M_i. In order to avoid establishing an obvious relationship between M_i and l_i, Alice uses an encryption algorithm to encrypt M_i and get ciphertext c_i. According to predefined mapping rules, c_i is mapped into label l_i separately. After converting secret message into attributes, Alice inputs l_i and I^i_{cover} into N_H, so the cover image I_{cover} is translated into stego image I_{stego} with l. Finally, Alice transmits I_{stego} through public channel and wishes Bob can extract M correctly.

As mentioned in Section II-D, among candidates for multi-domain image translation we use StarGAN in MDI to realize the generation of stego image. The details of the generator G and discriminator D in StarGAN are introduced in [14]. During training, StarGAN is directly fed by original image and target label of multi-domain, uses a single generator to learn how to translate an original image into the corresponding domain by generating a random target domain label. By doing this, StarGAN can translate original image into any domain. Hence we use this advantage to map M into target label l_i and obtain a I^i_{stego}. And the more attributes are selected during training, the more secret message a stego image I_{stego} can contain.

Specifically, the input of G is an original image and a target label. The output of G is an generated image that belongs to target label. During training G, target label vector firstly

Algorithm 1 The Algorithm of Stego Image Generation

Input:
  M: secret message.
  $AllAttr$: all of attributes in the image.
  L_{attr}: the fixed length of selected attributes in the N_H
  Map: the map relation between message segment and attributes label.
  I_{cover}: the set of cover images.
  key: the secret key used to encrypt.
Output: The set of stego images I_{stego}.
Algorithm:
Select appropriate attributes $Attr$ = Select($AllAttr$)
N_H = Train(I_{cover}, $Attr$)
Compute the length of M as $L_M = Length(M)$
$n = \lceil L_M / L_{attr} \rceil$
Divide M into n segments from 1 to n
for index $i \in$ range(0, n) **do**
  $m = M_i$
  if ($i == n$) **then**
    padding m with $L_M - (n-1) \times L_{attr}$ 0
  end if
  c_i = Encrypt(m)
  $l_i = Map(c_i)$
end for
for index $i \in$ range(0, n) **do**
  $I^i_{stego} = N_H(I^i_{cover}, l_i)$
end for
return I_{stego}

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replicates spatially and concatenates with image vector, then they are used as the input of G. The output of G needs to be input to D and D distinguishes the real/fake of generated image and the attribute labels. At the same time, this generated image needs to be input to G again with the label of original image to maintain the cycle consistency with the help of L1 norm. When training D, the input is only real image. The input of D is a real/fake image and the output is two labels that they are a real/fake label and a predicted attributes label. So D can not only distinguish if an image is real or fake, but also classify the attributes of an image, which is different from common function of discriminator in GAN.

In image translation, though it's possible for GAN to translate any attributes in the image, the generated image may have a low visual effect. Hence in MDI, some attributes in the image are not suitable for translation to utilize. As mentioned in [21], it's difficult for traditional metrics to evaluate the quality of generated images. So how to determine which attributes are appropriate to translate is a problem. Inspired by [21] and [14], firstly Alice uses all attributes to train a GAN and generates corresponding target images. Secondly, Alice also trains a classifier with all attributes in real image dataset. Thirdly, she uses this classifier to measure the generated image with target label as a pseudo-metric. By doing this, Alice can clear which attributes are suitable. The consideration is that if the generated images with target label are realistic, classifiers trained on corresponding attributes label will be able to classify the attributes of synthesized image correctly as well. However, there may be some conflicting attributes here, Alice can filter them by herself and obtains selected attributes at last.

What's more, aiming to confuse the relationship between secret message and target label, before mapping message into binary label, Alice would encrypt M_i at first. In order to avoid the situation where Bob gets multi-bits message extraction error because of one bit ciphertext extraction error due to the existence of image attack in the public channel, MDI adopt stream cipher which uses only a single bit as the basic encryption unit and has limited error propagation.

C. Secret message Extraction

In the stage of extraction there are also four steps. In the first step, Alice uses N_H to obtain some generated images. In the second step, Alice trains a classifier as N_E fed by generated images and shares N_E with Bob. In the third step, Bob uses N_E to extract the label l' in I'_{stego}. At last, Bob decrypts c'_i into M'_i and splices these segments into a whole secret message M'.

The algorithm for secret message extraction is shown in Algorithm 2. To generate the training set of N_E, Alice synthesizes images by N_H. Then she trains a classifier N_E which is used to classify the attributes of images and shares it with Bob. Bob can use N_E to take in ${I^i_{stego}}'$ as input and gets l' from I'_{stego}. Next Bob uses the mapping rules to inverse map l' into encrypted message segments c'_i. At last because Alice uses encryption algorithm to secure the message, Bob should use the secret key to decrypt c'_i into M'_i. At last all M'_i are spliced into M'. After M' has been spliced, a complete communication is over.

Algorithm 2 The Algorithm of Secret message Extraction

Input:
N_H: Network for Hiding.
I_{cover}: the set of cover images.
I'_{stego}: the set of stego images with noise.
Map: the map relation between message segment and attributes label.
key: the secret key used to encrypt.
Output: Recovered secret message M'.
Algorithm:
Obtain training dataset with data augmentation I_{train} = Data_Augmentation($N_H(I_{cover})$)
N_E = Train(I_{train})
Compute the number of I'_{stego} as $n = Length(I'_{stego})$
for index $i \in range(0, n)$ **do**
  $l'_i = N_E(I'_{stego_i})$
  c'_i = Map(l'_i)
  M'_i = Decrypt(c'_i, key)
end for
return M'

TABLE II
TOP-15 PERFORMANCE OF ATTRIBUTES TRANSLATION. HIGHER IS BATTER FOR ACCURACY.

Attribute	Accuracy(%)
Mouth Slightly Open	96.60
Narrow Eyes	95.60
Smiling	94.80
Black Hair	92.00
Gray Hair	86.00
Pale Skin	84.60
Brown Hair	84.20
Blond Hair	84.00
Bangs	82.80
Chubby	81.60
Eyeglasses	80.00
Heavy Makeup	73.60
Male	70.00
Double Chin	63.80
Bald	63.00

When extracting binary attribute labels of stego images, N_E can output the label of an image. We tend to choose a multi-label classifier as N_E compared with binary classifier. Because stego images have undergone multi-domain translation, using a multi-label classifier is more efficient and lightweight, instead of a series of binary classifiers. We use ResNet-152 as the backbone of multi-label classifier N_E then connect a fully connected layer to output the classification result. The output dimension of fully connected layer is equal to L_{attr}.

For making the model resist a variety of image attack and be more robust, MDI adopts data augmentation on N_E. Alice use N_H to translate some original images into target images. Then she attacks these images to add disturbance. In this paper, Alice will try all the common types of image attack to generate

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attacked images. The types and parameters of image attacks used in this paper can be seen in Section IV-C. When training N_E, if the number of target images and attack types are m and n respectively, there are $m \times (n+1)$ images in the training set totally. In the mini-batch for each iteration, a single type of image are fed into N_E and a new type of image are input in next iteration. Empirically, after training with attacked images, N_E can be robust to common types of image attack. We will demonstrate it with an ablation study in Section IV-C.

IV. Experiments and discussion

In this section, we compare MDI with several coverless steganography methods as our baseline, i.e. S-DCGAN, proposed by Hu et al. [7], S-DRAGAN [8], S-StarGAN [13], SIFT-BOF [9] and SIFT-HASH [4].

The results of experiment focus on two aspects: robustness and hiding capacity. We will also discuss the security and coverage rate of our framework.

A. Setup

In our experiments, MDI ran on a machine with Intel Xeon E5-2620 2.10GHz CPU, 16GB memory, 12GB TITAN Xp GPU, Ubuntu 18.04 LTS and PyTorch 1.4.0 [23]. The dataset chosen in our method is celebA (CelebFaces Attributes) [24]. This dataset contains 202,599 face images of celebrities with 178×218 resolution, and each image is annotated with 40 binary attributes, e.g. arched eyebrows, big lips.

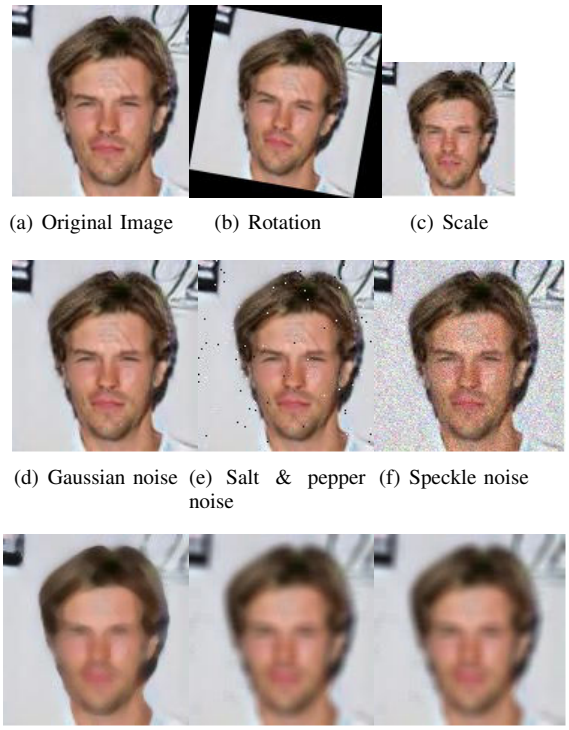
The training of StarGAN in MDI used Adam [25] with β_1 = 0.5 and β_2 = 0.999, learning rate of 0.0001 for the first 10 epochs, linearly decay the learning rate to 0 over the next 10 epochs and performed one generator update after five discriminator updates. Above all are the same with StarGAN [14].

We wish our comparison result is fair. So for S-DCGAN, we used the code provided by [7] and adjusted the model to reach a good performance. Besides TensorFlow 1.3.0 [26] was required, the running environment and dataset was the same as MDI. For S-StarGAN, we trained a StarGAN again for comparison using the same parameters mentioned in [13]. The selected attributes is the same with us. There is no available source code in SIFT-BOF and SIFT-HASH, so we completed these two methods by Python 3.6 according to the details in corresponding papers. The dataset used for these methods is CelebA dataset too. S-DRAGAN, there is no source code and as mentioned in [8] the dataset was collected and optimized by themselves. Hence we just used their results in paper directly.

B. Analysis of Attributes Selection

We trained and tested a StarGAN model which can translate all the 40 attributes in celebA. The generated target images only change one attributes from original image. Then we used multi-label classifier trained with celebA having accuracy of 92% to score attributes by the classification accuracy. The results of top-15 are shown in TABLE II.

But there are still some conflicting attributes here, e.g. bald and hair color. Hence according to the actual situation, we finally chose 11 of them as the attributes used to image steganography, which can realize a good transformation visual effect, such as black hair, smile, gender and so on.



(a) Original Image (b) Rotation (c) Scale

(d) Gaussian noise (e) Salt & pepper noise (f) Speckle noise

(g) Median filter (h) Mean filter (i) Gaussian filter

Fig. 3. The different visual effect of different image attack.

C. Analysis of Robustness

Robustness is an important evaluation indicator in image steganography. In public channel, there are attackers carrying out image attack on stego images to modify pixel value, image size, or even high-level semantics. If the secret message is bound to these features, it is inevitable for Bob to introduce error when recovering secret message. Hence if an steganography framework has strong robustness, the secret message will be recovered correctly to the most extent. There are several image attacks mentioned in [27] as follows, and we will use them for robustness experiments:

1) Rotation, with angles of 10°, 30°and 50°;
2) Scale, with ratios of 0.3, 0.5, 0.75, 1.5 and 3;
3) Gaussian noise, with variance of 0.001, 0.005 and 0.1;
4) Salt & pepper noise, with variance of 0.001 and 0.005;
5) Speckle noise, with variance of 0.01, 0.05 and 0.1;
6) Median filter, with size of 3 × 3, 5 × 5 and 7× 7;
7) Mean filter, with size of 3 × 3, 5 × 5 and 7× 7;
8) Gaussian filter, with size of 3 × 3, 5 × 5 and 7× 7.

In order to vividly illustrate the visual effect of above attacks, the images under attack are shown in Fig. 3. As a result, I_{stego} sent by Alice has changed into stego images with disturbance I'_{stego}, which will have an impact on Bob's extraction.

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TABLE III
COMPARATIVE STUDY ON ROBUST BETWEEN DIFFERENT METHODS(%).

Attack	S-DCGAN	S-DRAGAN	S-StarGAN	SIFT-BOF	SIFT-HASH	MDI without Data Augmentation	MDI
Rotation 10°	6.24	**90.63**	61.32	51.25	51.64	68.90	86.25
Rotation 30°	6.22	**88.10**	59.41	49.25	51.68	60.66	79.41
Rotation 50°	6.17	**83.24**	59.48	50.90	52.23	60.68	78.18
Scale 0.3	6.09	10.08	64.01	49.38	52.94	76.41	**87.89**
Scale 0.5	6.13	41.08	65.91	50.13	53.00	80.93	**91.55**
Scale 0.75	6.17	76.46	65.61	48.92	53.04	80.57	**91.93**
Scale 1.5	6.22	**94.18**	65.73	51.13	54.79	81.00	92.36
Scale 3	6.25	**94.95**	65.56	50.60	53.77	81.00	92.55
Gaussian Noise 0.001	6.20	84.52	65.69	49.82	52.83	81.05	**92.66**
Gaussian Noise 0.005	6.09	53.20	64.39	48.93	52.42	80.42	**92.03**
Gaussian Noise 0.1	5.99	3.62	63.87	46.78	51.76	79.89	**91.57**
Salt & Pepper Noise 0.001	6.13	83.91	65.69	50.15	52.77	81.11	**92.61**
Salt & Pepper Noise 0.005	6.18	50.96	65.60	49.02	51.90	80.66	**92.20**
Speckle Noise 0.01	6.20	91.28	65.66	50.73	53.56	81.14	**92.65**
Speckle Noise 0.05	6.22	76.59	65.62	50.03	52.93	81.36	**92.11**
Speckle Noise 0.1	6.19	8.93	65.49	49.30	52.16	81.41	**92.20**
Median Filter 3 × 3	6.19	81.87	65.10	50.05	52.92	80.05	**92.36**
Median Filter 5 × 5	6.26	65.14	64.21	50.15	52.17	78.55	**90.89**
Median Filter 7 × 7	6.28	53.01	63.32	50.83	51.26	76.02	**89.25**
Mean Filter 3 × 3	6.24	75.76	64.91	49.50	52.72	79.93	**91.84**
Mean Filter 5 × 5	6.25	81.39	64.00	49.32	51.82	77.82	**90.77**
Mean Filter 7 × 7	6.30	29.76	62.89	50.85	51.40	74.39	**88.91**
Gaussian Filter 3 × 3	6.24	82.57	65.11	50.33	51.54	80.20	**92.16**
Gaussian Filter 5 × 5	6.23	69.49	64.69	50.57	52.21	79.80	**91.66**
Gaussian Filter 7 × 7	6.20	50.16	64.11	50.10	51.58	78.34	**91.09**

In our paper, we use bit correct rate (BCR), computed as shown in (1) to measure the robustness of these methods,

$$BCR = \frac{\sum_{i=0}^{Length(M')} \sum_{j=0}^{Length(M'_i)} sim(M_{ij}, M'_{ij})}{\sum_{i=0}^{Length(M')} Length(M'_i)} \quad (1)$$

where $Length(M')$ means the number of M', $Length(M'_i)$ means the number of bits contained in each M'_i. The function $sim(x, y)$ is used to judge the similarity of bit x and y, if same return 1, otherwise return 0.

As mentioned in Section III-C, we used 500 generated target images and added 25 image attacks with different parameters. So there are 13,000 images as training set and validation set to train a N_E. The ratio of training set to verification set is 5:1. In order to ensure the authenticity of the experimental results, we used 7,500 unseen stego images with attack as the test set of MDI. In practice, some of bits may be used as error correction code to improve the accuracy of message extraction, but in this paper, instead of error correction code, all bits are used to transmit the message.

The experiment results are shown in Table III. S-DCGAN has a worse performance because when the noise is added into stego image, it no longer corresponds to the initial noise vector. So extractor cannot recover the secret message correctly. Under rotation attack and parts of scale attack, S-DRAGAN has a better performance than ours. Although S-StarGAN also uses StarGAN, it has a low BCR after all. On the contrary, MDI is the best one among these five methods under other attacks.

In order to prove the effectiveness of data augmentation, we also design an ablation study. The training set of ablation study only consist of 1000 generated images without attack. The result is shown in next to last column in Table III. All of the BCR has been improved. So we attribute our performance to data augmentation when training N_E.

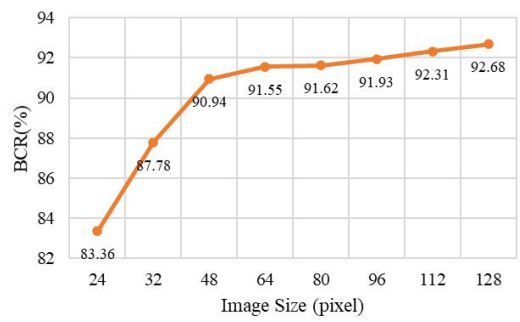

Fig. 4. With the decrease of image size, BCR also decreases.

It should be noted that center crop attack is not used in this work. Because S-DRAGAN, S-StarGAN and MDI both put the characters or face in the center of the whole image. If there is a center crop attack, it will destroy the high-level semantic information. So we see it as a malicious attack and do not consider it in this paper.

D. Analysis of Hiding Capacity

The capacity of stego images depends on the size (pixel × pixel) of stego image and the length of the label each image contains. In this paper, we use bit per pixel (bpp) shown as

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TABLE IV
THE HIDING CAPACITY BETWEEN DIFFERENT METHODS.

	Length of code (bit)	Size of image (pixel × pixel)	hiding capacity (bpp)
S-DCGAN	37.5	64×64	9.16×10^{-3}
S-DRAGAN	14	128×128	8.54×10^{-4}
S-StarGAN	5	64×64	1.22×10^{-3}
SIFT-BOF	8	512×512	3.05×10^{-5}
SIFT-HASH	18	512×512	6.86×10^{-5}
MDI	11	32×32	$\mathbf{1.07\times10^{-2}}$

(2) as the evaluation indicator of hiding capacity,

$$bpp = \frac{Length(label)}{W \times H} \tag{2}$$

where $Length(label)$ is the length of the label, W is the width of the image, and H is the height of the image. The less the size of the image is, the more hiding capacity is.

In previous subsection, we found the size of image also influences the BCR of extracting secret messages. So we use an acceptable reduction (about 10%) in BCR in exchange for an increase in hiding capacity. According to Fig. 4, we scale image size to 32 × 32. In this case the hiding capacity can be achieved as 1.07 $\times10^{-2}$ bpp, which can be seen in the last row of Table IV. The second column is the length of the code (bit), the third column is the size (pixel × pixel) of the image and the last column is hiding capacity (bpp). Column 2-4 show the capacities of GAN based methods. Column 5-6 show the capacities of traditional methods. So our method outperforms state-of-the-art methods.

E. Discussion of Security

We assume there are two analyzers, denoted as Eve-I and Eve-II in the public channel. Eve-I attempts to reveal the label according to the stego image and Eve-II is a common steganalyzer. In particular, Eve-I attempts to reveal the label according to the stego image. He can train a similar model to classify the stego images and naturally link secret message and label together as long as he refers the possible binary attributes selected by Alice and Bob. Eve-II attempts to distinguish if there is secret message in stego image.

In order to defend Eve-I, we adopt stream cipher in MDI. With stream cipher algorithm, besides one bit error in ciphertext will not introduce into other bits when deciphering, encryption algorithm will also make the relationship between secret message and image attribute label more confused. What's more, cover images set can be kept as a secret in real application. Without cover image set, it makes attacker more difficult to know which attributes are selected to generate stego image.

For the Eve-II, coverless image steganography icoverage rates naturally resistant to steganalysis. We used PSRM [28] to extract the feature of stego image and ensemble classifier proposed by Fridrich et al. [29] to train and classify cover/stego images. We use Err as evaluation indicator, as shown in (3)

$$Err = (1 - \frac{TN + TP}{ALL}) \tag{3}$$

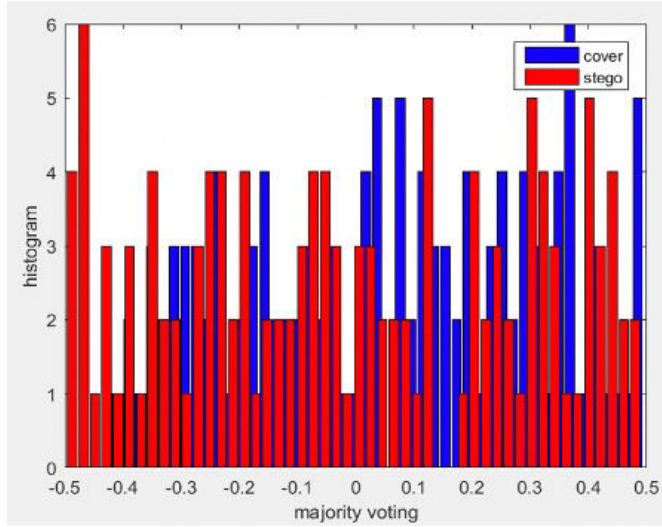

Fig. 5. The feature distribution comparison between cover images and stego images.

where TP and TN means the number of no secret message and having secret message predicted correctly separately, All donates the number of all the images. The PSRM feature distribution of cover image and stego image is shown as Fig. 5 and the result is 50.20%. It means Eve-II cannot distinguish if there is a secret message in the image.

F. Discussion of Coverage Rate

Finally, we will discuss the coverage rate of MDI. The notion of coverage rate is proposed by [5], which is shown as (4)

$$P = \frac{Num}{M} \times 100\% \tag{4}$$

where Num is the number of secret message that can be represented in the cover image set, M is the number of secret message that need to be represented.

MDI bases on multi-domain image translation so a certain image can be converted into different domain image according to different target labels. Utilizing MDI, Alice even just needs only one image to generate all the images that secret message maps. Therefore MDI can achieve 100% coverage rate.

V. CONCLUSION

In this paper, we proposed MDI, a coverless image steganography framework based on multi-domain image translation to establish a relationship between secret message and generated image. With the help of GAN, MDI can realize 100% coverage rate. Based on multi-domain image translation, MDI can have a higher hiding capacity. In addition, an extra multi-label classifier with data augmentation can improve the robust of the framework. Our experiment results demonstrate MDI performs state-of-the-arts in robustness and hiding capacity.

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VI. Acknowledgment

The authors would like to particularly acknowledge our supervisor, Bin Wu, for his patient support and advice during this research.

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